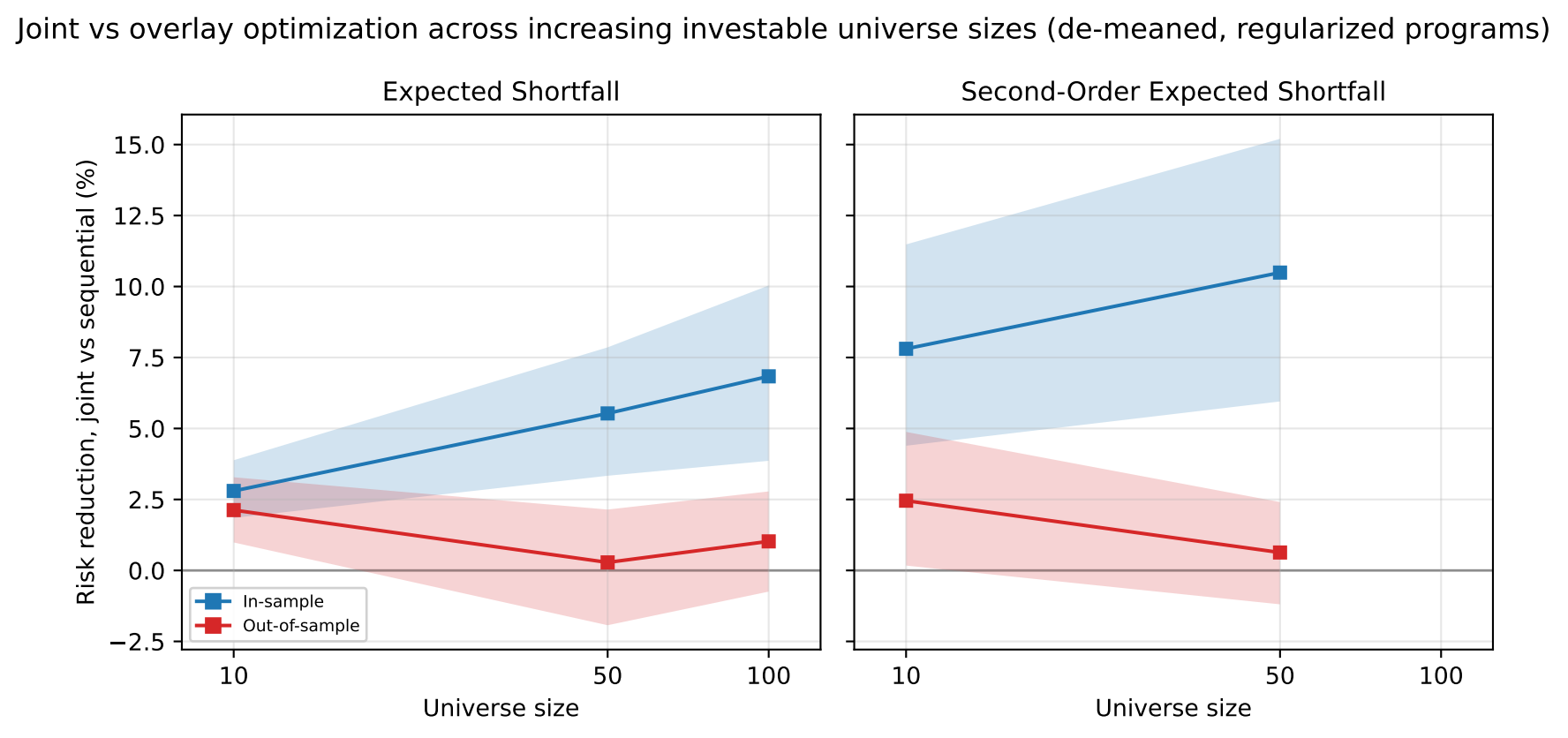
Joint vs overlay optimization across increasing investable universe sizes (de-meaned, regularized programs)
Expected Shortfall
Second-Order Expected Shortfall
Risk reduction, joint vs sequential (%)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
In-sample
Out-of-sample
10
50
100
Universe size